

HERNANDO, PANTINE B.

BSIT-3

1-MONTH NETOWRK FORENSICS (FEBRUARY)

WEEK 3 - Applied Network Forensics Challenges

1. Objective

The goal of this week's activity was to reinforce network forensics concepts by analyzing PCAP datasets using **Wireshark**, identifying protocols, reconstructing streams, detecting abnormal activity, and extracting transferred data.

2. Tool Used

- **Wireshark** – for packet capture analysis, protocol identification, stream reconstruction, and data extraction

3. Methodology

- I. Opened the PCAP dataset in Wireshark and applied protocol filters (TCP, UDP, HTTP, DNS) to isolate relevant traffic.
- II. Used "Follow TCP/UDP Stream" to reconstruct full sessions and examined HTTP requests/responses for transferred files or credentials.
- III. Monitored traffic for unusual ports, abnormal patterns, and repeated failed connections.
- IV. Extracted files directly from HTTP streams and recorded any sensitive data visible in payloads.
- V. Re-applied different filters in Wireshark to ensure no suspicious activity was missed and documented findings systematically.

4. Findings

Protocol	Suspicious IP / DOMAIN	Transferred	Notes
HTTP	192.168.1.10	File download (example.txt)	Verified by following HTTP stream
DNS	suspicious.com	Query requests	Potential C2 activity
TCP	10.0.0.5	Login credentials (plaintext)	Found in TCP payload

- Observations: Unusual DNS queries and high-numbered TCP ports indicated suspicious activity. HTTP streams revealed file transfers inconsistent with normal usage.

5. Summary

PCAP analysis using Wireshark allowed identification of suspicious traffic, reconstruction of TCP/UDP streams, detection of abnormal activity, and extraction of data. Key findings included suspicious HTTP file transfers, unusual DNS queries, and potential sensitive data in TCP streams.

6. Reflection

This activity strengthened my skills in structured network forensic analysis using Wireshark. It improved my ability to detect protocol abuse, trace suspicious activity, and document findings efficiently while gaining confidence in interpreting network traffic.